

Daily Review

Booklet p.8-9 — Space Exploration and the ISS

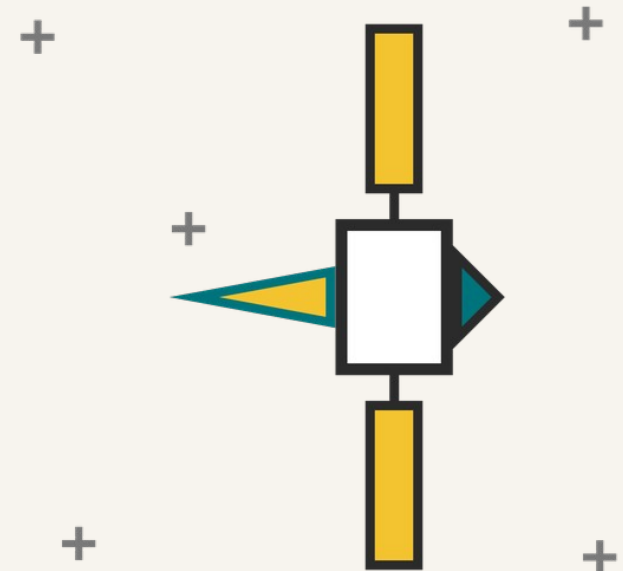
- 1 Name two risks of crewed space exploration.
- 2 Why does a craft far from the Sun need more cells?
- 3 What energy transfer happens in a solar cell?
- 4 What is a pressure differential?
- 5 What protects a craft on re-entry?

SPACE

Getting Around in Space

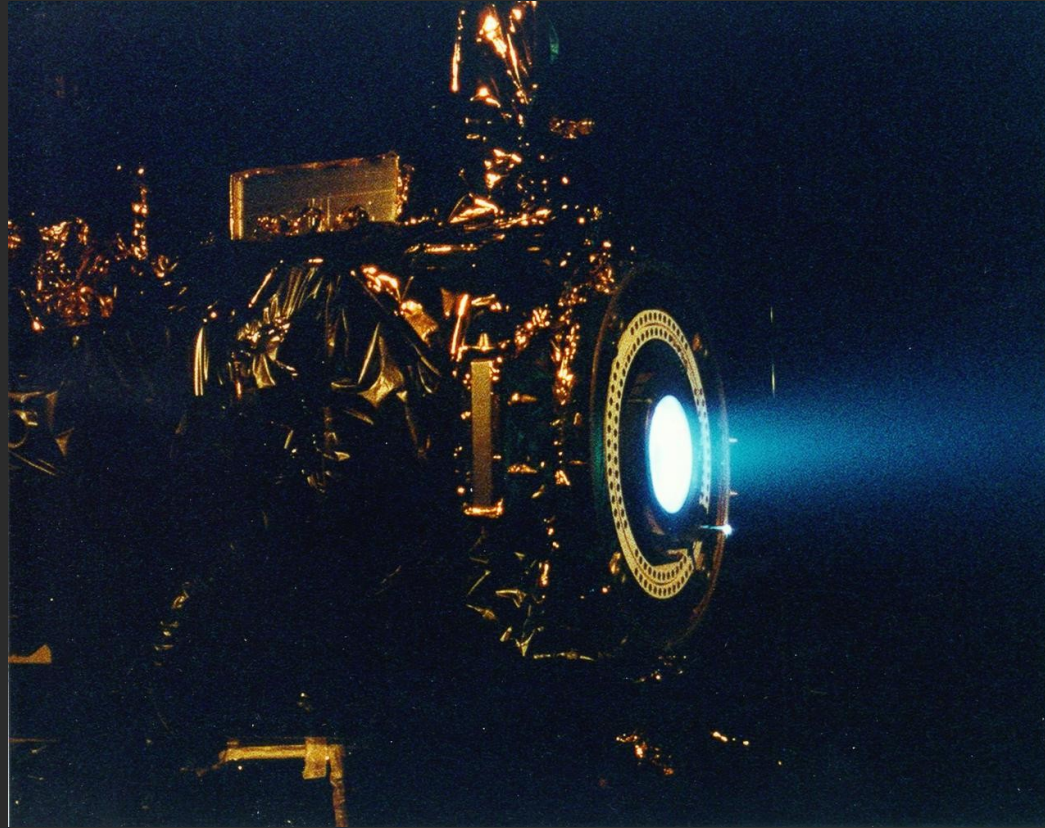
No air, no friction, no brakes.

National 5 Physics (S3)



THE BIG QUESTION

This engine's push is about the weight of a sheet of paper.
How can it be the fastest way to the outer Solar System?



The Deep Space 1 ion engine on test — NASA/JPL

PART 1

Moving and steering

How do you speed up, slow down and steer with nothing to push on?

By the end of this part...

- 1 I can describe the difficulties of manoeuvring in deep space
- 2 I can explain how a spacecraft changes speed and direction

Show me — Describe each word

SHOW ME BOARDS

1

friction

2

thrust

3

balanced forces

4

unbalanced force

5

deep space

Show me — Describe each word

SHOW ME BOARDS

6 manoeuvre

7 docking

8 ion drive

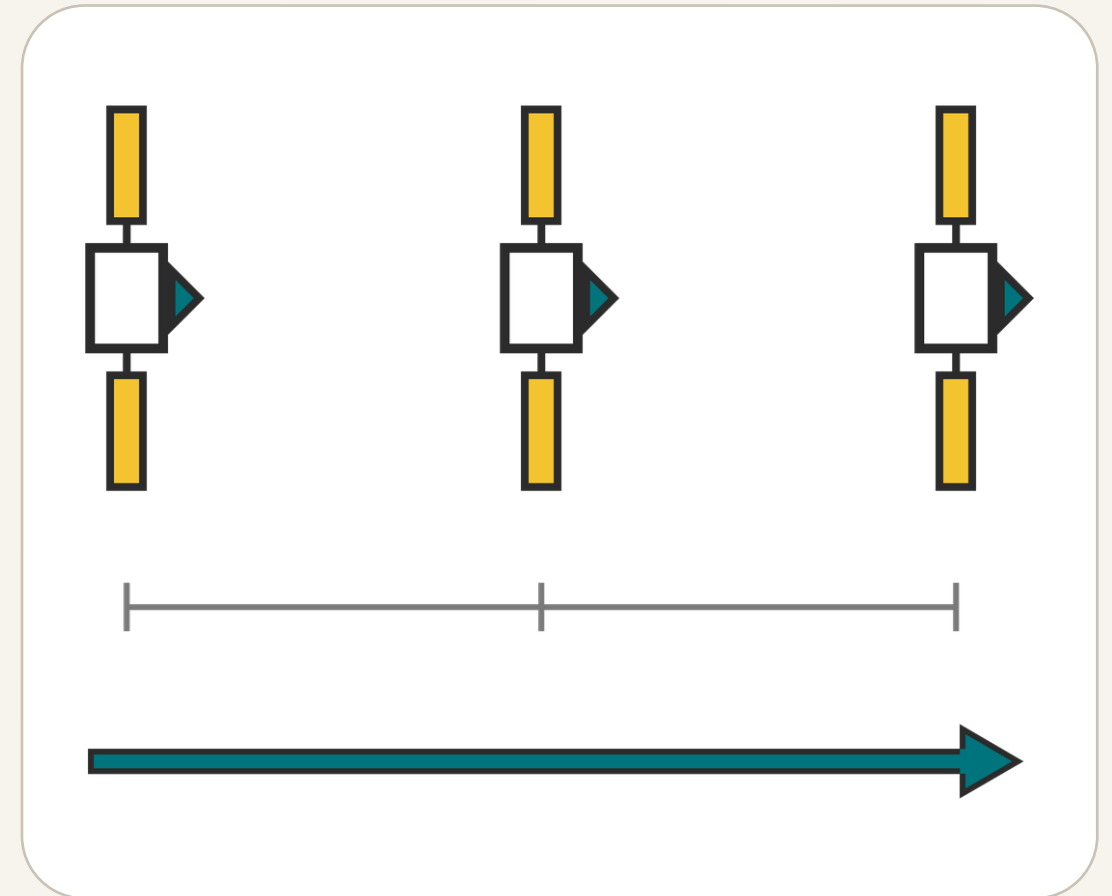
9 catapult

10 fuel

No friction in deep space

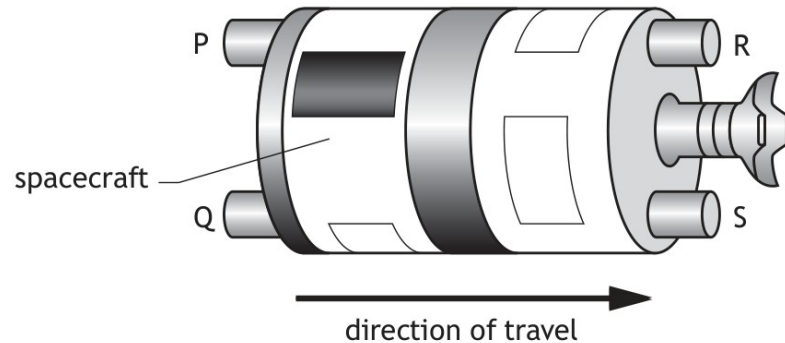
Booklet p.10

- Far out in space there is no air, so no friction.
- Engine off: the forces on the craft are balanced.
- It keeps moving at a constant speed in a straight line.



Show me – which engines?

SHOW ME BOARDS



- 1 Which two engines slow the craft down?
- 2 Which two engines speed it up?
- 3 All engines off. Describe its motion.
- 4 Why does it not slow down on its own?

Docking with the ISS

Booklet p.10, Tasks 1 and 2



- Both orbit at 7.7 km s^{-1} .
- Only the difference in their speeds matters.
- Tiny thruster pushes close the last few metres.
- With no friction, every push must be cancelled.

Show me – moving in space

SHOW ME BOARDS

EXTRA PRACTICE

- 1 A probe switches its engine off in deep space. What happens?
- 2 Why is there no friction in deep space?
- 3 How does a spacecraft slow down?
- 4 A craft starts to spin. How can it stop spinning?
- 5 Why is docking so difficult?

Tasks 1 and 2 – dock with the ISS

Booklet p.10

Try to dock with the ISS in the simulator.
Then describe why it was difficult.

Booklet p.10, Tasks 1 and 2



[Docking simulator](#)

Task 2 – Answers

ANSWERS

Booklet p.10

- There is no friction, so once it moves, it keeps moving.
- Every push must be cancelled by a push the other way.
- Small thrusters give tiny changes, so it takes a long time.
- The ISS and your craft are both moving very fast in orbit.

Daily Review

Booklet p.10 — Part 1

- 1 Why is there no friction in deep space?
- 2 Engine off in deep space: what happens?
- 3 How does a spacecraft slow down?
- 4 What makes docking with the ISS hard?
- 5 What is thrust?

PART 2

Getting there faster

Two ways to cross the Solar System — and neither is brute force.

By the end of this part...

- 1 I can describe the purpose and function of the catapult
- 2 I can describe the purpose and function of an ion drive

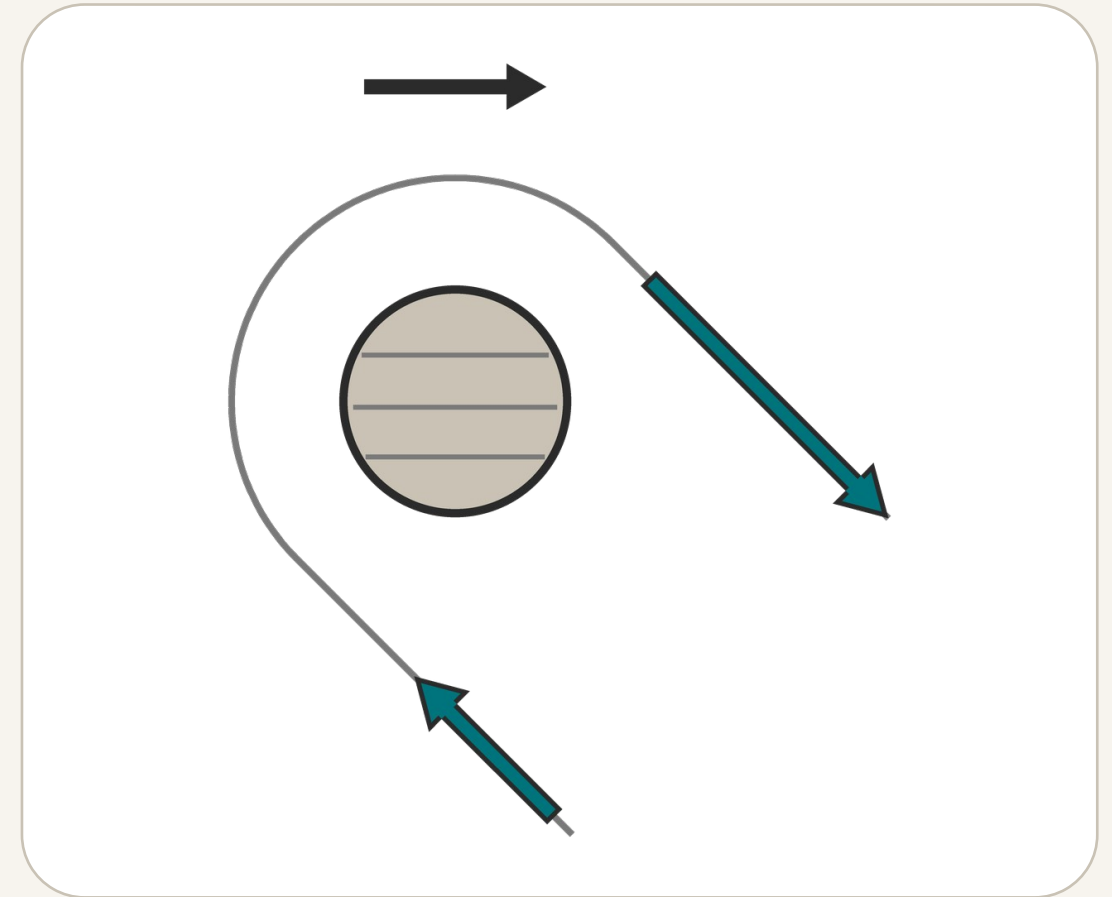


Jupiter — NASA Science / Voyager and Cassini missions

The catapult

Booklet p.10, Task 3

- A craft is steered close to a moving planet or moon.
- Its gravity pulls the craft round and speeds it up.
- It gains speed without burning fuel.
- Artemis II used the Moon's gravity to curve its path home.



Task 3 – the catapult

Booklet p.10

Watch the clip about the gravity assist.
Then describe it in your own words.

Booklet p.10, Task 3

Show me – the catapult

SHOW ME BOARDS

- 1 What is another name for the catapult manoeuvre?
- 2 What pulls the craft towards the planet?
- 3 What happens to the craft's speed?
- 4 Why does this save fuel?
- 5 Name an object a craft could catapult from.

Show me – the catapult

SHOW ME BOARDS

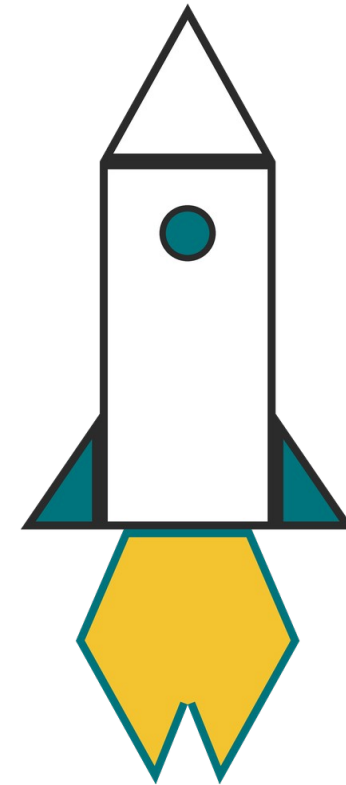
EXTRA PRACTICE

- 1 Does the catapult need the engine to be on?
- 2 What changes as well as the craft's speed?
- 3 Why does a mission to Neptune use a catapult?
- 4 Why must the craft be steered very accurately?
- 5 What if it passed too close?

Chemical rockets

Booklet p.10, Task 4

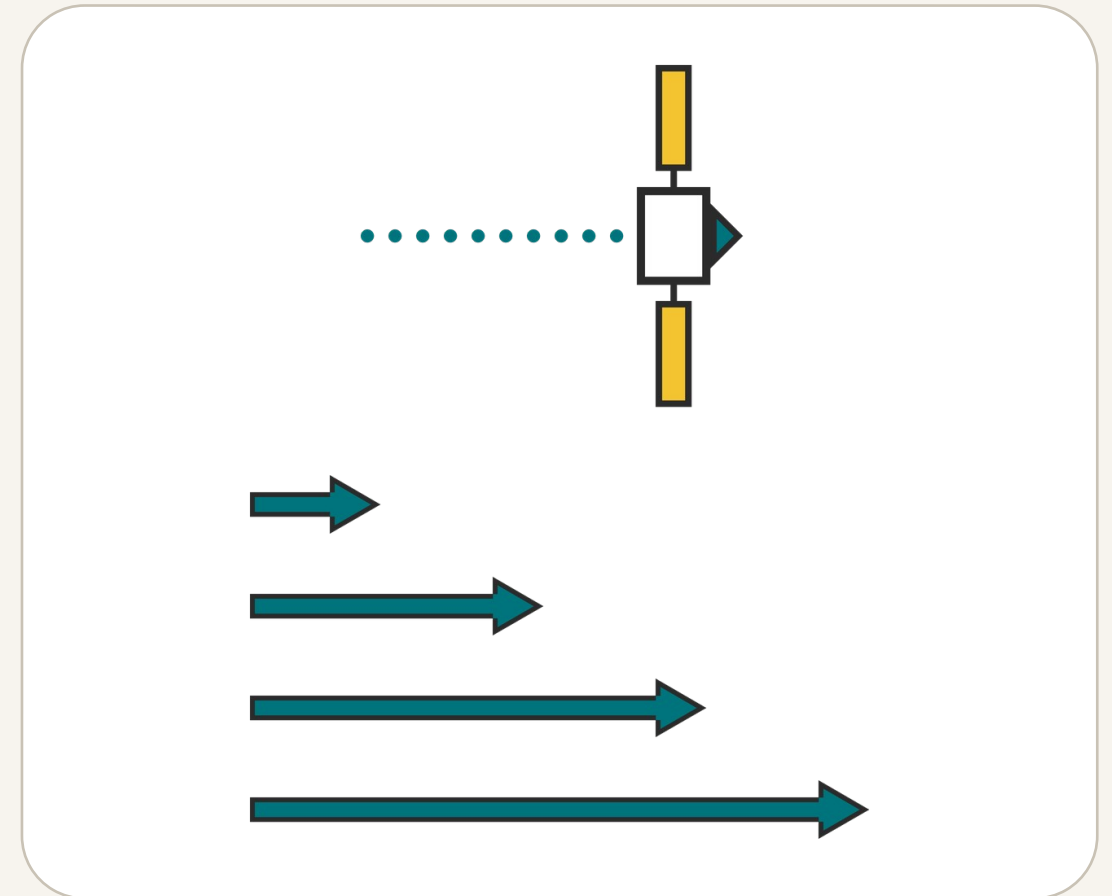
- Burning fuel pushes hot gas out very fast.
- A huge thrust, but only for a few minutes.
- Lift-off needs a thrust bigger than the weight.



The ion drive

Booklet p.10, Task 4

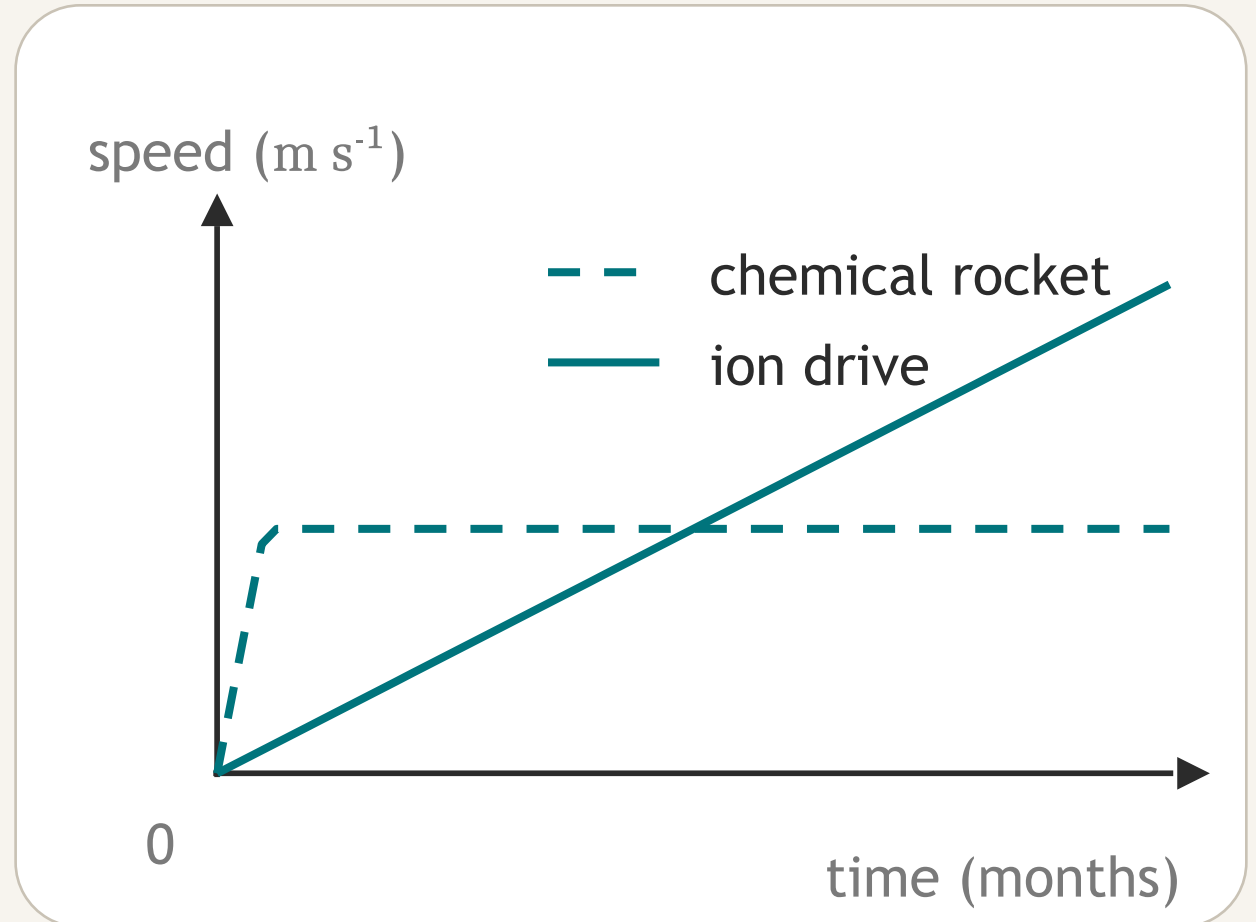
- Pushes out charged particles, using very little fuel.
- A small unbalanced force, over a very long time.
- So the craft reaches a very high speed in the end.



Why the ion drive wins in the end

Same journey, two engines

- The ion drive's force is tiny, so a is tiny.
- But it fires for months, so the speed keeps climbing.
- The rocket burns out in minutes, then coasts.
- Given time, the ion drive ends up far faster.



Agree, disagree or it depends?

SHOW ME BOARDS

Boards up: AGREE, DISAGREE or IT DEPENDS — and why

- 1 A spacecraft needs its engine on to keep moving.
- 2 There is no gravity in deep space.
- 3 An ion drive is more powerful than a rocket.
- 4 The catapult speeds a craft up without fuel.
- 5 You cannot steer without air.

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

An ion drive's thrust is only about 0.1 N.
The main reason it still reaches a very high speed is

- A** it has no mass to accelerate
- B** it keeps pushing for months on end
- C** its thrust grows as it travels
- D** there is no gravity in deep space
- E** it carries more fuel than a rocket.

Show me – engines

SHOW ME BOARDS

EXTRA PRACTICE

- 1 What must be true of the thrust at lift-off?
- 2 Why are ion drives used for long journeys?
- 3 Why is a small force useful in deep space?
- 4 Give one drawback of an ion drive.
- 5 Which engine uses fuel faster?

Task 4 – types of rocket engine

Booklet p.10

Fill in the table for a chemical rocket
and for an ion drive.

Booklet p.10, Task 4

Task 4 – Answers

ANSWERS

Booklet p.10

Engine	Description	Benefits	Drawbacks
Chemical	burns fuel; hot gas pushed out	huge thrust; can lift off	uses fuel very fast
Ion drive	pushes out charged particles	very little fuel; high top speed	tiny thrust; cannot lift off

Would you rather...

TALK IT OUT

...send a probe to Jupiter on an ion drive,
...or catapult it past Venus and Earth first?

Which gets there sooner? Which uses less fuel?
What could go wrong with each?

Extension — space in everyday life

Booklet p.11

Find out how space research has
changed things we use every day.

Booklet p.11, Extension



[NASA at home](#)

Checkpoint

- 1 Why does a craft in deep space keep moving with its engine off?
- 2 How can an ion drive's tiny force give a very high speed?
- 3 Describe the catapult manoeuvre, and say why it is used.

Exam practice

Past paper

- National 5 Physics 2025, Section 2, question 4(a). 2 marks.
- Question 4(b)(i). 1 mark. A probe on its way to an asteroid.
- Then booklet p.11, Task 5.

LOOKING AHEAD

Satellites

What do satellites do for us — and why does height set the orbit?

Plenary

- 1 The big question: how can a tiny push be the fastest way out there?
- 2 Why is there no friction in deep space?
- 3 Engine off, deep space: describe the motion.
- 4 Describe how an ion drive works.
- 5 Describe the catapult manoeuvre.

Plenary

EXTRA PRACTICE

- 1 How does a spacecraft change direction?
- 2 What makes docking with the ISS difficult?
- 3 Give one benefit of an ion drive.
- 4 What does the catapult change about the motion?
- 5 Which engine gives the bigger thrust?

Extension

- 1 Explain why a probe's engine can be off for years of its journey.
- 2 Suggest why a craft uses both a chemical rocket and an ion drive.
- 3 Suggest why a catapult cannot be used on every part of a journey.

Extension

EXTRA PRACTICE

- 1 A thruster fires for two seconds to start a spin. How is it stopped?
- 2 Why would an ion drive be useless in Earth's atmosphere?
- 3 Suggest why a probe to Saturn might pass Venus first.

Course specification – 1 of 3

N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the challenges of space travel:
 - travelling large distances with the possible solution of attaining high velocity by using ion drive (producing a small unbalanced force over an extended period of time)
 - travelling large distances using a ‘catapult’ from a fast moving asteroid, moon or planet

Course specification – 2 of 3

N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the challenges of space travel:
 - manoeuvring a spacecraft in a zero friction environment, possibly to dock with the ISS
 - maintaining sufficient energy to operate life support systems in a spacecraft, with the possible solution of using solar cells with area that varies with distance from the Sun

Energy for life support was taught last lesson.

Course specification — 3 of 3

N5 Physics course specification v4.0, p.6 — Newton's laws

- Application of Newton's laws and balanced forces to explain constant velocity (or speed), making reference to frictional forces.

In space the frictional force is zero.